

**NATIONAL UNIVERSITY OF MODERN
LANGUAGE ISLAMABAD**

Project report



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Project Report: Voting System:

1. Project Overview

The provided Java project implements a **Voting System** that allows:

- **Admin functionalities:** Adding voters and candidates.
- **Voter functionalities:** casting votes.
- **Result viewing:** Displaying election results to all users.

The project leverages a Java Swing GUI for a user-friendly interface and organizes its features into various components for maintainability.

2. Key Features

1. User Interfaces:

- **Main Window:** Entry point for admin and voters.
- **Admin Login & Dashboard:** Secure admin access to manage voters and candidates.
- **Voter Login:** Allows registered voters to log in and cast their votes.
- **Results Window:** Displays election results in a tabular format.

2. Core Functionality:

- **Registration:** Handles voter and candidate registration with unique identifiers.
- **Authentication:** Validates admin credentials.
- **Voting Logic:** Ensures each voter can only vote once, and updates vote counts accordingly.

3. GUI Components:

- Attractive Swing components (e.g., JFrame, JLabel, JButton) with custom styling.
 - **GridBagLayout** for flexible component arrangement.
 - Dynamic updates to data displayed in tables and dropdown menus.
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3. Structure & Code Organization

The project is divided into multiple files for modularity:

- **Main Application** (Main.java): Launches the application and initializes data.
- **GUI Windows:**
 - MainWindow.java
 - AdminLoginWindow.java, AdminDashboardWindow.java
 - AddVoterWindow.java, AddCandidateWindow.java

- ResultsWindow.java, VotingWindow.java
 - **Data Classes:**
 - Voter: Represents a voter with ID, name, and voting status.
 - Candidate: Represents a candidate with ID, name, and vote count.
 - **Logic:**
 - Registration: Handles storage and addition of voters and candidates.
 - Admin: Provides hardcoded admin credentials and authentication logic.
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4. Highlights

- **Dynamic Data Handling:**
 - Candidate and voter details are dynamically managed in ArrayLists.
 - Results and voting options update based on runtime data.
 - **Error Handling:**
 - Prevents duplicate IDs for candidates and voters.
 - Ensures invalid credentials and already-used votes are handled gracefully.
 - **User Experience:**
 - Well-designed GUI with styled components for clarity.
 - Confirmation and error dialogs to guide user actions.
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5. GitHub Repository

The full project code is hosted on GitHub. You can access it at the following link:

[GitHub Repository: Voting System](#)

6. Test cases:

Test cases for voter login

ID	Input	description	Expected output	Actual output	Class	Status
Case-01	123	Correct ID	Successful login	Successful login	Valid	Pass
Case-02	12345	Incorrect ID	Failed	Failed	valid	fail
Case-03	123	Already voted	Failed	Failed	valid	fail

Test cases for adding a new voter

ID	Input	description	Expected output	Actual output	Class	Status
Case-01	Name: ali ID: 123 Area: f-10	Inter a new voter details	Successful added	Successful added	Valid	Pass
Case-02	Name: ali ID: 123 Area: f-10	Inter the details of an existing voter	Failed	Failed	valid	pass
Case-03	Name: hassan ID: 123 Area: g-7	Inter new voter details but with an existing voter ID	Failed	Failed	valid	pass

7. Conclusion

The project is a well-structured implementation of a basic voting system with clear objectives and a user-friendly interface. It demonstrates excellent use of Java Swing for GUI design and object-oriented programming principles for modularity. With data persistence and improved security, the system could be scaled for practical use.